Roll No. .....

MAI–205

M. B. A. (AI & DS) (SECOND SEMESTER)
END SEMESTER EXAMINATION, July/Aug., 2022

MACHINE LEARNING USING R

Time : Three Hours

Maximum Marks : 100

Note : (i) This question paper contains two Sections-Section A and B

(ii) Both Sections are compulsory.

(iii) Answer any *two* sub-questions among (a), (b) and (c) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section—A

1. (a) What is machine learning ? Briefly explain its applications and uses. (CO2)

(b) Briefly explain the steps involved in building a Machine Learning application. (CO2)

(c) Differentiate between Classification supervised learning and Regression supervised learning. (CO4)

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2. (a) What are the different data types in R ? Briefly explain about them. (CO2)

   (b) Explain Confusion Matrix with a well suitable example. (CO2)

   (c) Explain the difference between Linear Regression and Logistic Regression. Give some examples of each. (CO4)

3. (a) Explain Naïve Bayes classifier with some examples. (CO2)

   (b) Describe Dimensionality Reduction and its types. Give few examples. (CO2)

   (c) Explain any *two* algorithms that justify regression supervised learning. (CO5)

4. (a) What is Correlation ? Discuss the uses of Correlation techniques. (CO2)

   (b) Explain PCA (Principal Component Analysis). (CO2)

   (c) Draw all the plots available in R and state their uses. Write one line command for it. (CO2)

Section—B

5. **Case Study :** (CO5)

   R is a tool for statistical analysis, machine learning and visualization. R interprets the code and makes the development of code easier. Answer below mentioned questions regarding R language.

   (1) How do you install a package in R?

   (2) z < - 0/0. What is the class of z ?

   (3) Which function is used to view the dataset in spreadsheet like format ?

   (4) How do you typecast a dataframe into a matrix ?

   (5) Difference between vector and factor.

(6) A homogeneous 1-D and 2-D data structure in R is called an atomic vector and matrix respectively. What is the name for :

(a) a 1-D heterogeneous data structure,

(b) a 2-D heterogeneous data structure ?

(7) Function of decompose command in time series forecasting.

(8) Write the command to read read a csv file with 5 rows in a dataframe.

(9) Give one difference between summary() and describe() command in R.